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Optimizing Speaker Identification through SincsquareNet and SincNet Fusion with Attention Mechanism

Banala Saritha[a,*], Rabul Hussain Laskar [a],Madhuchhanda Choudhury[a],Anish Monsley K[b]

[a]*Department of Electronics and Communication Engineering, National Institute of Technology Silchar, Assam, 788010, India*
[b]*Department of Applied Mechanics and Biomedical Engineering, Indian Institute of Technology, Madras, 600036, India.*

banalasarita@gmail.com*, rhlaskar1@ece.nits.ac.in, m_chanda@ece.nits.ac.in, anishmonslay@yahoo.com

Abstract

Advancements in machine learning and deep learning benefit access control systems, forensics, and biometrics particularly in speaker identification systems. The SincNet architecture is a distinct convolutional neural network (CNN) designed for speaker identification. It operates by taking one-dimensional raw speech and directing it through its initial convolutional layer, consisting of Sinc filters. In this work, we present a SincsquareNet, a CNN that efficiently learns customized triangular band-pass filters using trainable Sinc-squared functions. Also, we propose the fusion of SincsqaureNet with SincNet for robust speaker identification. Further, a self-attention mechanism is employed to obtain discriminating features. The Librispeech dataset is used to validate the proposed framework. This approach makes use of the best of both filters, helps the network learn more robust features, and converges faster. The findings of the experiments show that, when compared to SincsquareNet, speaker identification accuracy was relatively improved by 8%, while validation loss was reduced by 7%.

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Keywords: Speaker identification, SincNet, SincsquareNet, Feature fusion, Attention mechanism.

1. Introduction

Human communication is primarily based on speech. In addition to conveying ideas and thoughts, speech captures significantly additional details, like gender identity, cognitive behavior, age, and speaker identity [1]. The technique of identifying a speaker from the acoustic properties of a particular utterance is known as speaker identification (SI). It is a vital field of research with numerous applications, including forensic speech verification

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for suspect identification, access control to various services, such as voice dialing, voice biometrics, secure voice based authentication systems, real-time speech processing, access to computer surveillance, mobile commerce, and mobile payments [2]. Deep neural networks (DNNs) have drawn significant interest in the field of speaker identification. The DNN had been trained before using handcrafted speech features. Due to the lossless nature, handling raw waveforms became an active research area. Deep learning methods outperform conventional approaches by extracting discriminative features from raw speech using high-level data abstraction. Ravanelli et al. [3] introduced SincNet as an efficient deep learning-centered approach for the speaker identification task. In this work, we presented SincsquareNet, a novel approach to the speaker identification task inspired by the SincNet architecture. However, the existing networks have the following shortcomings: i) they are ineffective at extracting more distinguishing features as they treat all features equally. ii) The speaker identification performance is subpar when compared to state-of-the-art systems (SOTA). iii) The computational complexity of the system increases due to varying dimensions of raw speech signals. To address these limitations, this research work introduces the fusion technique and attention mechanism. These are the major contributions of this paper:

• Concatenation of features: Here we introduce the fusion of SincNet with SincsquareNet. This approach utilizes the advantages of both filters, making the network learn more useful features and converge faster.

• The attention mechanism is utilized to enhance the identification performance by concentrating on more discriminating features.

• The proposed method is used to extract significant features for further classification.

The organization of this paper is as follows: The associated work for dealing with raw speech signals is discussed in Section 2. The proposed fusion architecture is presented in Section 3. Section 4 describes the experimental design and findings. Finally, conclusion of the work was made in Section 5.

2. Related Works

Prior to the development of deep learning techniques, i-vector features were state-of-the-art for a long time for speaker identification tasks [4]. Existing approaches employed conventional feature extraction techniques including filter banks and Mel frequency cepstral coefficients (MFCCs). However, the temporal nature of speech is not retained by MFCC features because of their sensitivity to noise [5]. To mitigate this limitation, recent efforts have concentrated on providing the system with direct access to raw waveforms or spectrogram input. However, because of their rapid level variations, raw speech signals are typically more difficult to use than signals from other fields, like spectrograms and text. The advancements in deep learning made it easier to analyze the raw audio waveform using DNNs [6]. Georg Heigold et al. [7] proposed a d-vector and a b-vector for the extraction of speech features from raw speech and the identification of the speaker. These d-vectors were utilized by Jung et al. [8] to carry out speaker verification tasks. By emphasizing speaker discriminative data, Muckenhirn et al. [9] enhance speaker verification performance. To further enhance performance, Jung et al. [10] employed LSTM layers in the architecture and pre-emphasis on the preprocessing step. After extracting the frame-level features with residual blocks, Jung et al. [11] combined them using a gated recurrent unit (GRU) layer. SincNet, which interprets the raw speech input and extracts discriminant features, was introduced by Ravanelli et al. [12]. With the addition of Sinc filters to the first convolutional layer and a feature map scale, Jung et al. [13] further enhance the rawnet's efficacy.

In this work, we introduced SincsquareNet, which is a CNN that efficiently trains Sinc-squared functions to construct adapted triangular band-pass filters. The extraction of low-level features was the primary goal of the aforementioned works. Moreover, these techniques failed to acquire the distinctive features required for accurate speaker classification.

3. Proposed Method

This work presents a novel fusion of SincNet and SincsquareNet methods inspired by deep learning, which leverages robust features for speaker identification.

3.1. Feature Fusion Model

The proposed framework shown in Figure 1 includes a feature extraction module, a concatenated feature fusion layer with an attention mechanism module, and a classifier.

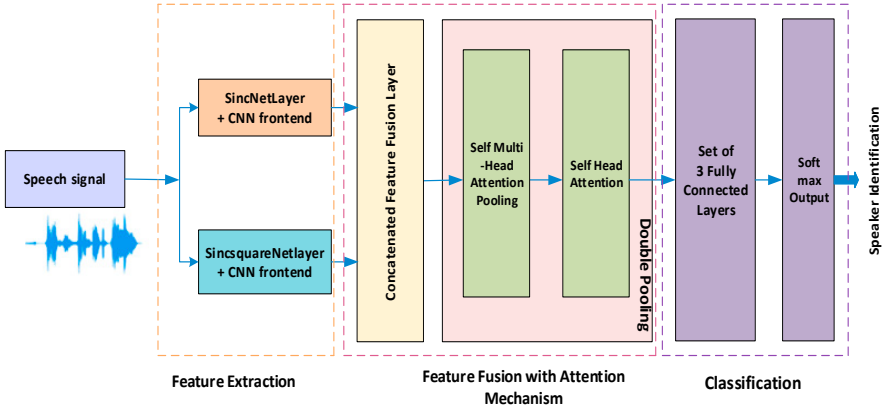


Figure 1.The framework of the proposed feature fusion model with attention mechanism

The SincNet and SincsquareNet simultaneously accept raw speech as input in feature extraction module. The extracted features from SincNet are concatenated with those from SincsquareNet in the fusion layer. Then double multi head attention is applied to the concatenated fusion layer to assign the weights to the concatenated features. The attention approach improves classification accuracy by focusing on the local information in the feature vectors. Further, these features are fed to classification module, consisting of three dense layers that are fully connected with Softmax as output layer. Leaky ReLU serves as the activation layer across all dense layers, while the classification layer is the Softmax layer.

3.2. SincNet Architecture

One of the most advanced deep learning-based architectures for speaker identification tasks that work with raw speech is SincNet. The Sinc functions in the first layer are convolved with the speech waveform to develop rectangular band-pass filters for extracting the low-level features [14]. The subsequent layers of CNN process these features further. The architecture of SincNet is shown in Figure 2. SincNet performs the convolution with a predetermined function' *r* 'that is dependent on finite learnable factors *θ*.

$$\mathbf{y[n]} = \mathbf{x[n]} * \mathbf{r[n,\theta]} \tag{1}$$

To use a filter bank made up of rectangular bandpass filters, specify r in this manner. The rectangular bandpass filters with cutoff frequencies f1 and f2 in the frequency domain.

$$R\,[f, f1, f2] = rect\left(\frac{f}{2f2}\right) - rect\left(\frac{f}{2f1}\right) \tag{2}$$

Now, the predefined function r in the time domain is represented as

$$r\,[n, f_1, f_2] = 2f_2\,sinc\,(2\pi f_2 n) - 2f_1\,sinc\,(2\pi f_1 n) \tag{3}$$

The *sinc* function is described here, as $sinc(x) = \frac{\sin(x)}{x}$

The cutoff frequencies f1 and f2 can be used to initialize filters. We must confirm that f1 ≥ 0 and f2 ≥ f1, the former equation is provided with the variables as:

$$f\bar{}1 = |f1| \quad (4)$$

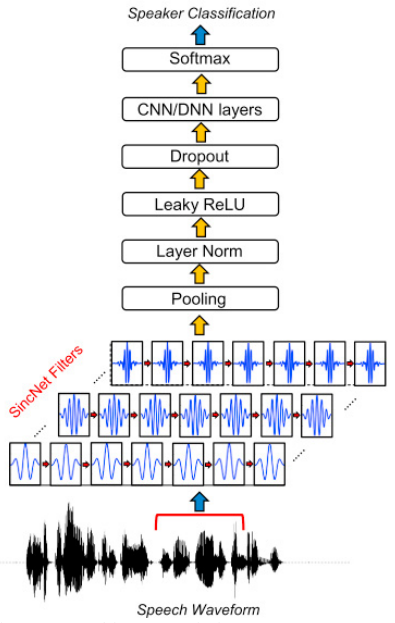


Figure 2. Architecture of SincNet [3]

$$f\bar{}2 = f1 + |f2 - f1| \quad (5)$$

$$f2abs = f1 + |f2 - f1| \quad (6)$$

Hamming windowing is performed by multiplying the function r by the window function $w_H(n)$ to soften the abrupt transitions.

$$rw_H[n, f\bar{}1, f\bar{}2] = r[n, f1, f2] \cdot w_H[n] \quad (7)$$

where $w_H(n)$ is given as

$$wH(n) = 0.54 - 0.46\cos\left(\frac{2\pi n}{K-1}\right) \quad n = 0, \ldots K-1 \quad (8)$$

3.3 SincsquareNet Architecture

We introduced SincsquareNet, a unique framework inspired by the SincNet architecture, designed for the task of speaker identification. Figure 3 illustrates the SincsquareNet architecture. This architecture efficiently learns customized triangular bandpass filters through trainable Sinc-squared functions [14]. The idea of human ear perception of sound motivates the use of triangular bandpass filters (TBPFs). We employed the Sincsquare function in the time domain to build the TBPFs in the frequency domain. These are similar to Mel filterbanks with lower frequencies being more discriminative and higher frequencies being less discriminative. The below equation demonstrates the operation of SincsquareNet where convolution is performed using a predefined parameter't' that relies solely on a limited set of adjustable variables:

$$\mathbf{y[n]} = \mathbf{x[n]} * \mathbf{t[n,\theta]} \quad (9)$$

To interpret t in in a manner compatible with TBPFs, an approach inspired by digital signal processing filtering is employed.

$$T[f, f1, f2] = tri\left(\frac{f}{2f2}\right) - tri\left(\frac{f}{2f1}\right) \quad (10)$$

Where f1 and f2 represent the obtained lower and higher cutoff frequencies, while 'tri (.)' signifies a triangular function within the frequency domain magnitude. Referencing function *t* in the time domain is expressed below when the inverse Fourier transform is applied [14].

$$t[n, f_1, f_2] = 2f_2\, sinc^2(2\pi f_2 n) - 2f_1\, sinc^2(2\pi f_1 n) \quad (11)$$

The Sincsquare function is written as $sinc^2(x) = \frac{sin^2(x)}{(x^2)}$

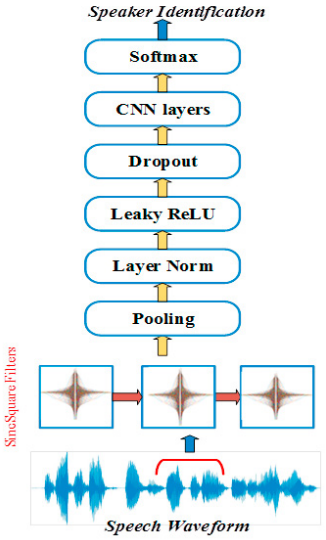


Figure 3. Architecture of SincsquareNet

The variables that accompany the lower and higher cutoff frequencies are as follows:

$$f\bar{}1 = |f1| \quad (12)$$

$$f\bar{}2 = f1 + |f2 - f1| \quad (13)$$

$$f2abs = f1 + |f2 - f1| \quad (14)$$

Next, the function t is multiplied by the window function $w_H(n)$ to apply hamming windowing.

$$t\, w_H[n, f\bar{}1, f\bar{}2] = t\,[n, f1, f2] \cdot w_H[n] \quad (15)$$

3.4 Double Multi-Head Attention Module

Research has shown that conventional neural network structures are unable to retain and analyze complex and large amounts of speech data. Giving such complex speech data to a CNN model for learning may result in the loss of essential information. Thus, it is essential to pay attention to the important information [15]. This can be accomplished by providing an attention mechanism to the CNN model. This research utilizes a double multi-head attention mechanism [16], where the pooling layer is augmented by a self-attention layer. It helps in obtaining more discriminative speech features compared to self-MHA pooling alone.

Let the sequence of hidden states from the concatenated features be denoted as $h = [h_1, h2....h_N]$. Assuming that m self-attention heads exists, as shown in Figure 4, and that the self-attention head computes the hidden state sequence $h_j=[h_{j1}, h_{j2} ...h_{jm}]$ is stated as follows:

$$c_j = \sum h_j^T w_j \quad (16)$$

Where c_j is the feature vector and w_j represents j^{th} attention head's weight.

Concatenating the vectors from all the heads gives the final feature vector, $c = [c_1 c_2... c_m]$. The double multi-head attention is determined as follows:

$$w_i' = \frac{exp(c_i^T u')}{\sum_{i=1}^{m} exp(c_i^T u')} \quad (17)$$

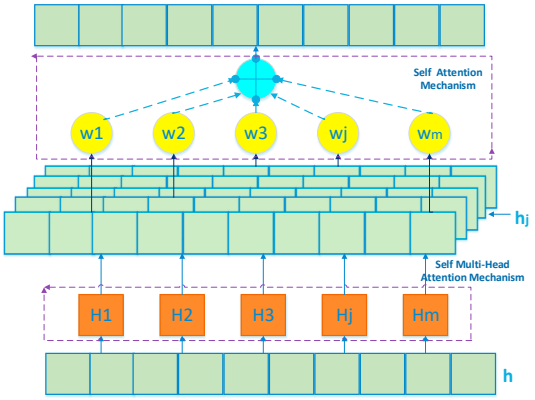


Figure 4. Double multi-head attention mechanism

$$c = \sum_{i=1}^{m} c_j^T w_i' \quad (18)$$

Where the w_i is the attention score of each head, in contrast to self-MHA pooling, c is the context vector obtained through the additional self-attention method from the weighted mean of the head context vectors.

3.5 Proposed Architecture

Figure 5 depicts the architecture of the proposed framework. The parallel implementations of SincNet and SincsquareNet form the feature extraction module. The SincNet architecture's first convolutional layer (conv1), named Sinclayer, comprises 80 rectangular band pass filters (RBPF) with 251 lengths in the frequency domain. Similarly, the first convolutional layer of Sincsquarenet is the Sincsquarelayer, which consists of 80 triangular band pass filters (TBPF) with 251 lengths.

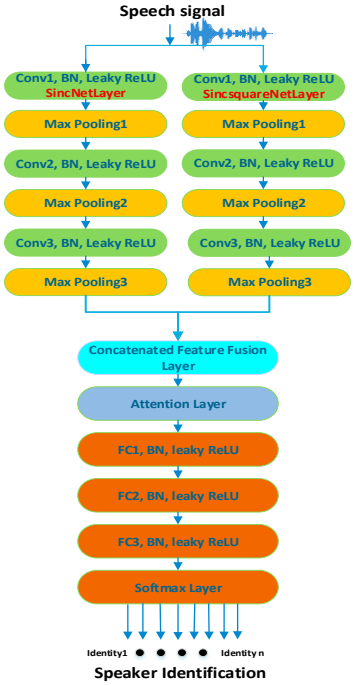


Figure 5. Architecture of the proposed model

Unlike the traditional CNN, the RBPF and TBPF learn only higher and lower cutoff frequencies during training so that they reduce the number of learnable parameters and extract more interpretable features. Next, both of these architectures include two convolutional layers (con2-conv3) with 60 filters of length 5. To speed up training and stabilize learning, layer normalization is added across all convolutional layers. Three max pooling layers (Max Pooling1- Max Pooling3) with pool size 1 and stride 3 are included to down sample the feature maps. The features from the two parallel networks are now concatenated in the feature fusion layer to produce more insightful and reliable features.

When compared to individual feature sets, these features are more robust. The primary benefit of feature-level fusion lies in its capability to identify interrelated features obtained by two parallel architectures, thereby discovering salient features for improving identification accuracy. Further, attention layer also helps in obtaining the salient features by assigning weights to the concatenated features. Double MHA layer helps in obtaining more discriminative speech features compared to self-MHA pooling alone. After that, 3 fully connected layers (FC1-FC3) with 256 neurons are normalized with batch normalization and leaky-ReLU nonlinearities. To improve overall performance and training stability Leaky-ReLU is preferred. Eventually a Softmax output layer is used as the classification layer.

This research examines six unique combinations of fusing techniques with an attention mechanism. 1. SincNet with self-attention 2. SincsquareNet with self-attention 3. Fusion of SincNet and SincsquareNet without any attention 4. Fusion of SincNet with SincsquareNet, where self-attention is applied to SincNet 5. Fusion of SincsquareNet with SincNet where self-attention is applied to SincNet 6. Fusion of SincNet and SincsquareNet with multi head attention. These experiments are summarized in Table 1.

4. Experiments & Results

To assess the proposed approach, experiments are conducted on well-known and publicly available Librispeech corpora. The experimental settings and database description are discussed in this section.

4.1 Database and Experimental Settings

The Librispeech database consists of 21933 samples from 2484 speakers and around 1000 hours of reading English speech acquired at a 16 kHz sampling rate [17]. Experiments are carried out on Librispeech corpora to assess the proposed method. To compare results, we used part of the Librispeech dataset that the authors of SincNet used, named train-clean-100. It is a gender-balanced database where 125 female and 126 male speakers are included with 100 hours of speech, and each speech is limited to 25 minutes per speaker. The database is divided into 15% test data, 15% validation data, and 70% speech files for training. Initially preprocessing of input is done by removing the silence regions. Every speech sample is split into 200 ms chunks with a 40 ms overlap after preprocessing. These chunks are fed to both the SincNet and SincsquareNet architectures in parallel. The utilization of a Softmax classifier facilitated the generation of frame-level speaker classifications, providing a range of posterior probabilities for each of the target speakers.

Training utilizes an Adam optimizer with specific configurations: lr = 0.001, $\alpha = 0.95$, $\epsilon = 10^{-5}$, and 128 minibatches. Hyper parameter tuning has been done on the Librispeech dataset to achieve optimal performance on both models. The experiments are conducted on a MATLAB platform running on a workstation equipped with an Intel Xeon Gold 6254 processor, 128 GB of RAM, and an NVIDIA Quadro P5000 graphics card (16 GB GPU).

4.2 Results and Discussion

To validate the efficacy of the presented model during training and testing, performance measures like speaker identification accuracy and loss are considered. The loss function represents the model's effectiveness, whereas accuracy indicates the percentage of correct predictions made by the model. Further, to assess the efficacy of the proposed framework, various investigations were carried out on the Librispeech database. These experimental combinations are discussed in section 3.5.

In this work, we trained the two convolutional neural networks, SincNet and SincsquareNet, with attention mechanisms and feature fusion techniques. SincsquareNet, our proposed approach, outperforms both standard CNN and SincNet. By employing a self-attention mechanism in both SincNet and SincsquareNet, we investigated E-I and E-II. Table 1 shows the experimental results of the presented framework.

Table 1. Experimental results of the proposed framework. Bold values represent best result.

Model	Modality	Final Validation Accuracy (%)	Final Validation Loss	Average Training Accuracy (%)	Average Training Loss	Testing Accuracy (%)
E-I	SincNet+Attention	80.23	0.783	82.31	0.739	80.23
E-II	SincsquareNet+Attention	83.31	0.741	84.88	0.702	83.31
E-III	SincNet+ SincsqaureNet	85.06	0.736	86.92	0.689	85.06
E-IV	E-I + SincsqaureNet	86.24	0.728	87.45	0.681	86.24
E-V	SincNet+E-II	87.85	0.719	88.84	0.673	87.85
E-VI	**E-III+Attention**	**88.93**	**0.689**	**90.06**	**0.649**	**88.93**

For easy analysis Figures 6 and 7 representing bar chart of information derived from Table 1. The comparison between the average training loss for the six experiments and the final validation loss is shown in Figure 6. The testing and average training accuracy for the six experiments are shown in Figure 7. A bar chart representing the information derived from Table 1, specifically depicting the validation loss and the mean training loss.

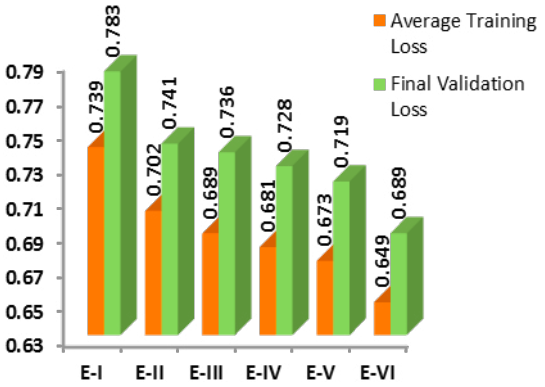


Figure 6. Comparison between training and validation loss.

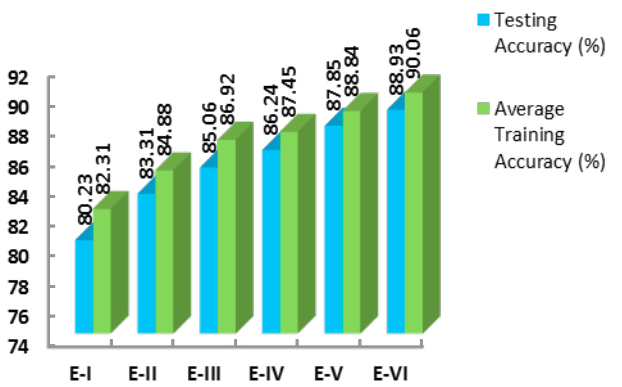


Figure 7. Comparison between training and testing accuracy

The identification accuracy improves by 3% from E-II to E-I. Due to the concatenated features, E-III surpasses E-II by 2%. The feature fusion technique, along with an attention mechanism, is employed in E-IV to E-VI. Furthermore, E-VI outperformed E-III by 4% and achieved the highest accuracy. This is because the discriminant features were improved using the double MHA method. Similar to this, the attention mechanism effectively reduced final validation loss by 5% from E-III to E-VI. The feature fusion and attention approaches collectively achieve the greatest loss reduction, about 27% compared to standard CNN.

Table 2 presents the speaker identification accuracy (SIA) of the proposed system along with a comparison to SOTA methods. SincsquareNet, our proposed work, improved SIA by 8% over standard CNN and by 4% over SincNet. With the highest improvement of 12%, the traditional fusion of SincNet and SincsquareNet (E-III) outperforms SOTA techniques. Eventually, in terms of SIA and loss, the proposed framework (E-VI) outperforms SOTA techniques. Also, it outperforms the fusion of Sinc and Sinsquare architectures with either of them using the attention mechanism.

Table 2. Comparative analysis of the proposed method with SOTA techniques. Bold values represent best result.

Model	Modality	SIA (%)
Standard CNN [3]	Standard CNN	73.58
SincNet [3]	SincNet	78.39
SincsquareNet	SincsquareNet	81.69
Ours - E-I	SincNet+Attention	80.23
Ours - E-II	SincsquareNet+Attention	**83.31**
Ours - E-III	SincNet+ SincsqaureNet	**85.06**
Ours - E-IV	E-I + SincsqaureNet	**86.24**
Ours - E-V	SincNet+E-II	**87.85**
Ours - E-VI	E-III+Attention	**88.93**

Table 3. Performance measures comparison with SOTA techniques.

Network	Accuracy	Precision	Recall	F-score
Standard CNN	0.7358	0.7342	0.7291	0.7291
SincNet	0.7839	0.7793	0.7702	0.7704
SincsquareNet	0.8169	0.8122	0.8112	0.8111
Proposed - E-VI	**0.8893**	**0.8798**	**0.8703**	**0.8702**

Figure 8 depicts a comparison of performance metrics for easy interpretation of the data in Table 3. Performance metrics show that concatenated features with attention mechanisms outperform other networks. The proposed framework achieved high levels of accuracy, F-score, precision, and recall. The results show that when we use feature concatenation, the model improves its correct predictions while decreasing its incorrect predictions.

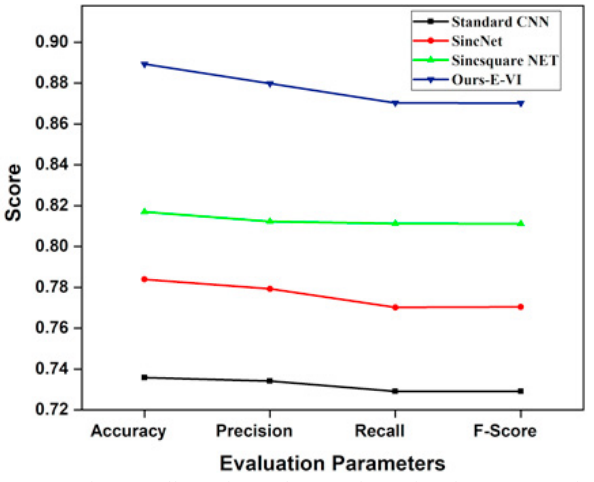


Figure 8. Illustration and comparison of performance metrics

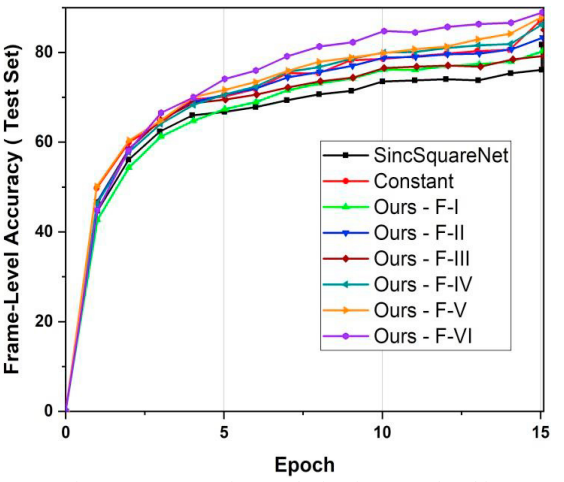


Figure 9. Comparative Analysis of proposed architectures in terms of accuracy

The test set's frame- level accuracy for a total of 15 epochs is shown in Figure 9. This graph illustrates the network's training effectiveness as the number of epochs increases.

5. Conclusion

In this work, we presented SincsquareNet, a DCNN based system designed for speaker identification directly from raw speech signals. It efficiently learns customized triangular band-pass filters using Sinc-squared functions. In

this research, we proposed the SincsqaureNet with a fusion of SincNet for robust speaker identification. Further, a self-attention mechanism is incorporated that focuses on informative segments and extracting the robust features. We validate the effectiveness of our proposed framework through extensive evaluations on the Librispeech database, using six different combinations of fusing techniques and an attention mechanism. The traditional fusion of SincNet and SincsquareNet outperforms state-of-the-art methods, with an improvement in speaker identification accuracy of 12%. The findings clearly indicate that E-VI modality consistently outperforms the fusion of SincNet and SincsquareNet, regardless of the attention mechanism employed with either of these architectures. The proposed method attains the highest accuracy of 88.93% in identifying the speakers through raw speech input. In the future, we will validate the effectiveness and robustness of our proposed system on a larger and more diverse dataset like Voxceleb. Additionally, an investigation of various loss functions will be conducted to further improve the performance of the system.

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